1. Title Page

Project Title: Smart Student Notes App
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Course: Flutter Development

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2. Introduction

Flutter is an open-source UI toolkit used for building cross-platform applications. In modern mobile applications, data storage plays an important role in saving user information locally.

This project demonstrates how to use local storage in Flutter using Hive database. The application allows users to manage notes efficiently without requiring an internet connection.

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3. Objective

The main objective of this project is:

- To develop a Flutter application using local storage
- To implement CRUD operations
- To ensure data persistence
- To create a user-friendly interface

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4. Tools & Technologies

- Flutter SDK
- Dart Programming Language
- Hive Database
- Android Studio

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5. System Requirements

Hardware:

- Minimum 4GB RAM
- Android Device / Emulator

Software:

- Android Studio
- Flutter SDK
- Git

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6. Project Workflow

The workflow of the application is as follows:

1. The user opens the application
2. The home screen displays saved notes
3. The user enters a note in the input field
4. The note is stored using Hive database
5. Notes are displayed using ListView
6. The user can edit an existing note
7. The updated note is saved in the database
8. The user can delete a note
9. The note is permanently removed from storage

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7. Working Explanation

- Hive database is initialized at app startup
- A storage box is created to store notes
- Data is added using `box.add()`
- Data is retrieved using `box.values`
- Data is updated using `box.putAt()`
- Data is deleted using `box.deleteAt()`

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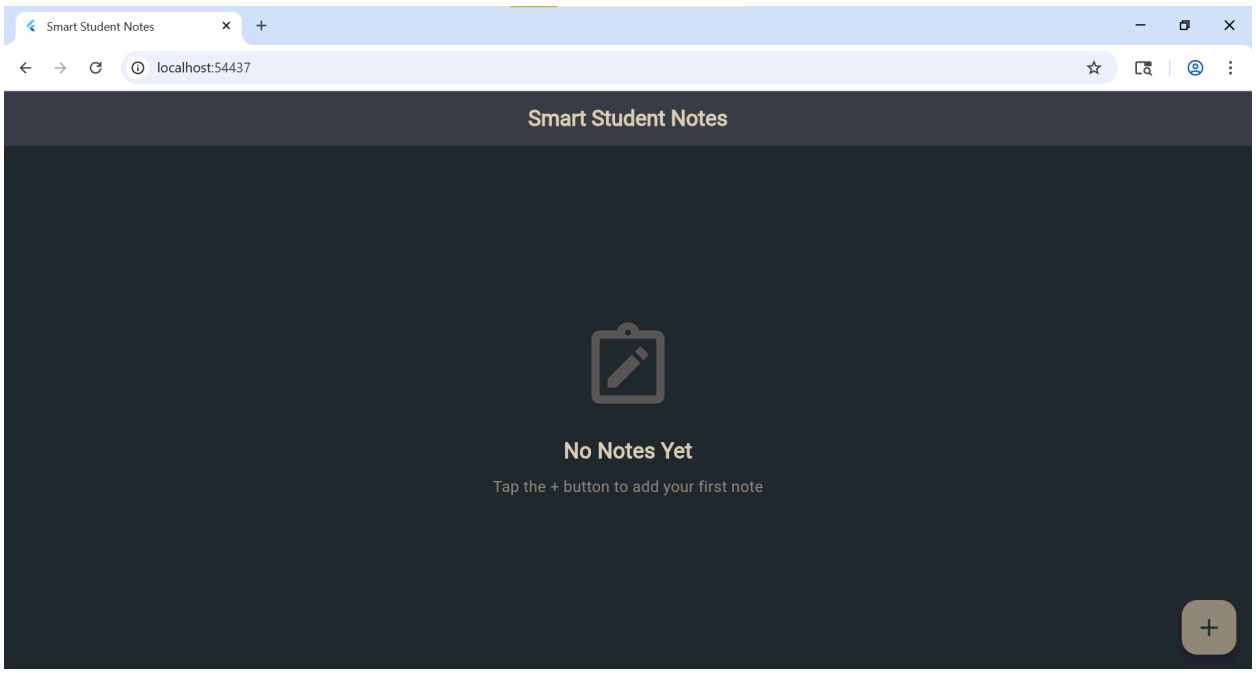
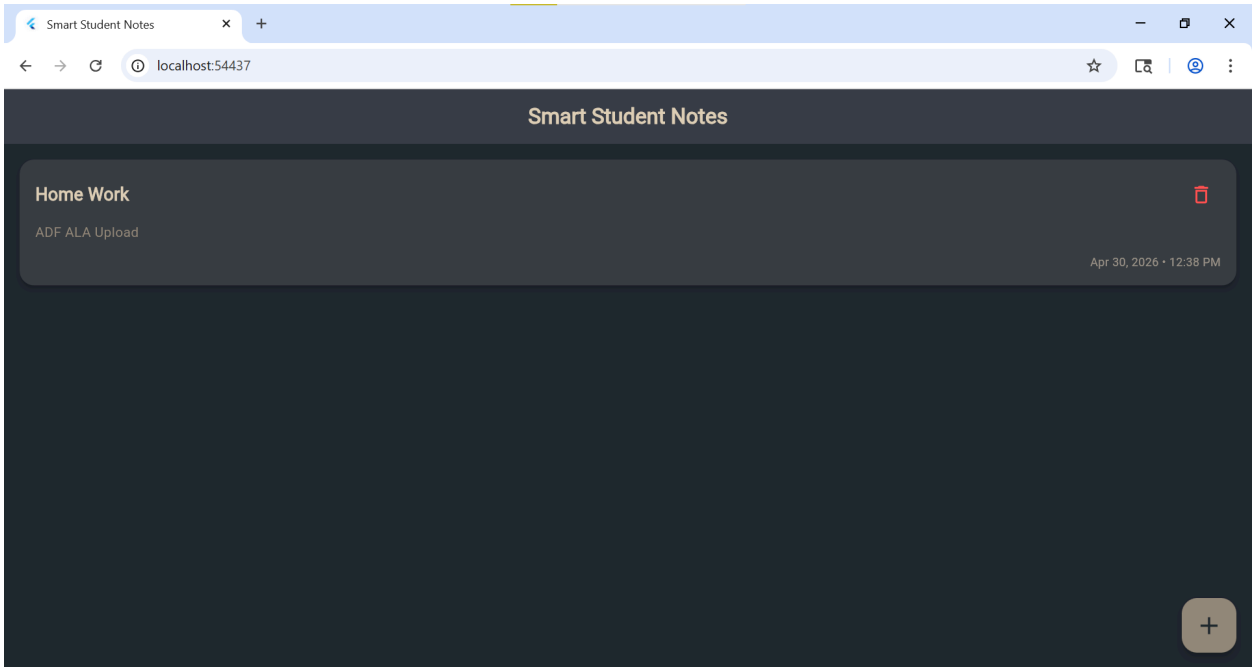
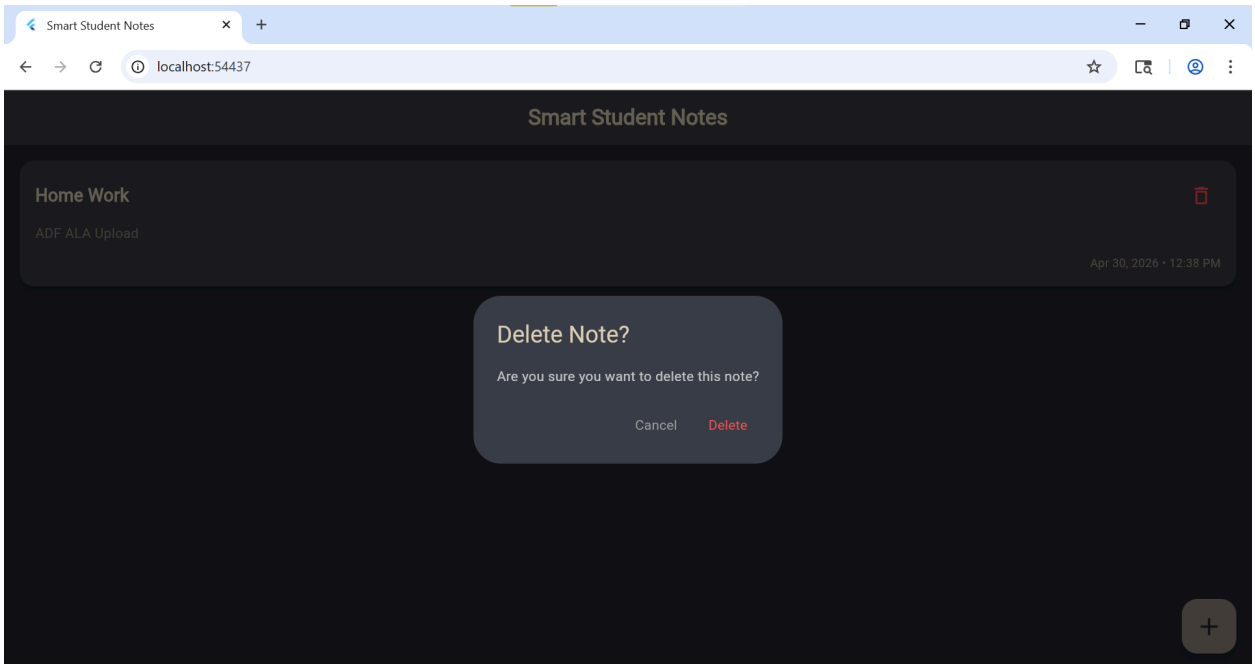
8. Screenshots

(Add screenshots of your app here)

- Home Screen
- 

- Add Note
- 

- Delete Note

- 

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9. Advantages

- Fast performance
- Offline data storage
- Easy to use
- Lightweight database

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10. Limitations

- Data stored only locally
- No cloud backup
- No user authentication

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11. Future Scope

- Add Firebase integration
- Implement login system
- Add reminders and notifications
- Add dark mode

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12. Conclusion

This project successfully demonstrates the implementation of local data storage in Flutter using Hive. It provides a simple and efficient way to perform CRUD operations and manage user data.

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13. References

- Flutter Documentation
- Hive Documentation
- Android Studio Guides